

DATA ANALYSIS INTERNSHIP

Task 9

Customer Analytics

Data Quality, Measures and Decision Support on 2,100 Customers

Data Cleaning • Measures and KPIs • Statistical Tests • Insights • Interactive Dashboard •
Quality Alerts

Prepared by

KOUAME Koffi Fidèle

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1 Executive summary

The brief asks for a complete cleaning, the relevant measures and KPIs, an interactive dashboard and clear insights, along the chain *raw data, clean data, measures, insights, decision support*. All are delivered, together with the notebook that reproduces every number and figure of this report and the pipeline (ETL, quality gate, scheduling and monitoring) that keeps the dashboard governed.

The extract holds **2,150 rows**. 50 are exact duplicates; once they are removed, **2,100 customers** remain, each with one purchase made between 27 October 2022 and 23 July 2025.

The customers spent \$1,021,324. The average ticket is **\$509.90** (median \$519.42), computed on the 2,003 recorded amounts; 97 amounts (4.6%) are missing. The mean satisfaction rating is **3.03 out of 5**, and 38.7% of the valid ratings are 1 or 2.

The primary finding concerns the data, not the customers. Four of the six quality policies are breached. Only 23.6% of customers have a usable age: the others carry the sentinels -1 or 200, or nothing. 291 ratings equal 10 on a 1 to 5 scale. 118 purchases carry the impossible date 32/13/2020. 565 purchases (26.9%) have no product category, more than any real category. The phone column holds two values for the whole file and 2,100 customers share 144 email addresses: no customer can be contacted from this register.

No customer attribute predicts spending. The amount does not differ by category ($p = 0.45$), by gender ($p = 0.10$), with age ($p = 0.02$) or with the rating ($p = -0.03$). Amounts are spread evenly between \$5 and \$1,000 ($p = 0.49$).

There is no seasonality. The dashboard's monthly chart shows a dip in August and September. It comes from coverage: those two months appear in two years of data, the others in three. On the 32 complete months, purchases are even ($p = 0.11$).

What the register supports well is the customer count, revenue, the average ticket and the satisfaction level. Targeting, contact and demographic analysis must wait for the capture to be repaired.

2 Data and method

2.1 The register

One row is one customer and one purchase. The columns are the customer identifier, name, email and phone, age and gender, the purchase amount, date and product category (Books, Clothing, Electronics, Home, Toys), and a satisfaction rating on the 1 to 5 scale. The file also carries two export artefacts: an empty column `Unnamed` and an exact copy of `Gender`.

2.2 Process

The analysis follows five steps, each implemented as a section of the notebook `Customer-Analytics.ipynb`:

1. **Quality audit.** Eight checks, one per defect family: duplicates, export artefacts, sentinel ages, out-of-scale ratings, the sentinel date, gender spellings and missing categories, contact fields, and the purchase amount.
2. **Cleaning.** Certain errors are corrected, uncertain values are flagged, and only exact duplicate rows are removed. The notebook checks that its cleaning gives the same table as the production pipeline `etl/run_etl.py`.
3. **Measures.** Each indicator is computed on the population that can support it, and that population is stated with the value.
4. **Tests.** Every segment difference is tested before it is reported: Kruskal–Wallis and Mann–Whitney for amounts across groups, chi-square for mixes and monthly counts, Spearman rank correlation for associations, Kolmogorov–Smirnov for the shape of the amount distribution.
5. **Decision support.** Each finding is translated into an action and an owner, exposed in an interactive dashboard and guarded by automated quality alerts.

The analysis uses Python only (pandas, NumPy, SciPy, Matplotlib). No spreadsheet and no business-intelligence tool is used at any stage.

3 Data quality

Table 1: Checks performed on the register, counted on the 2,100 customers after de-duplication

Check	Rows	Share
Rows as supplied	2,150	
Exact duplicate rows (repeated CustomerID)	50	2.3%
Customers after de-duplication	2,100	100%
Age missing or sentinel (−1, 200)	1,605	76.4%
Rating missing	322	15.3%
Rating out of scale (10)	291	13.9%
Product category missing	565	26.9%
Gender missing	267	12.7%
Purchase date invalid (32/13/2020)	118	5.6%
Purchase amount missing	97	4.6%
Phone missing or placeholder	2,100	100%

3.1 The defects are sentinels, not typing errors

Every defect of the file is a repeated default value. The 1,099 invalid ages are two values only, −1 (553) and 200 (546). The 291 out-of-scale ratings are all 10. The 118 invalid dates are all the same string, 32/13/2020. The phone column holds two numbers, 123456789 and 987654321, and is empty elsewhere. A defect repeated hundreds of times is a system default written when the real value was not captured; the fix therefore belongs to the capture forms, not to the analysis.

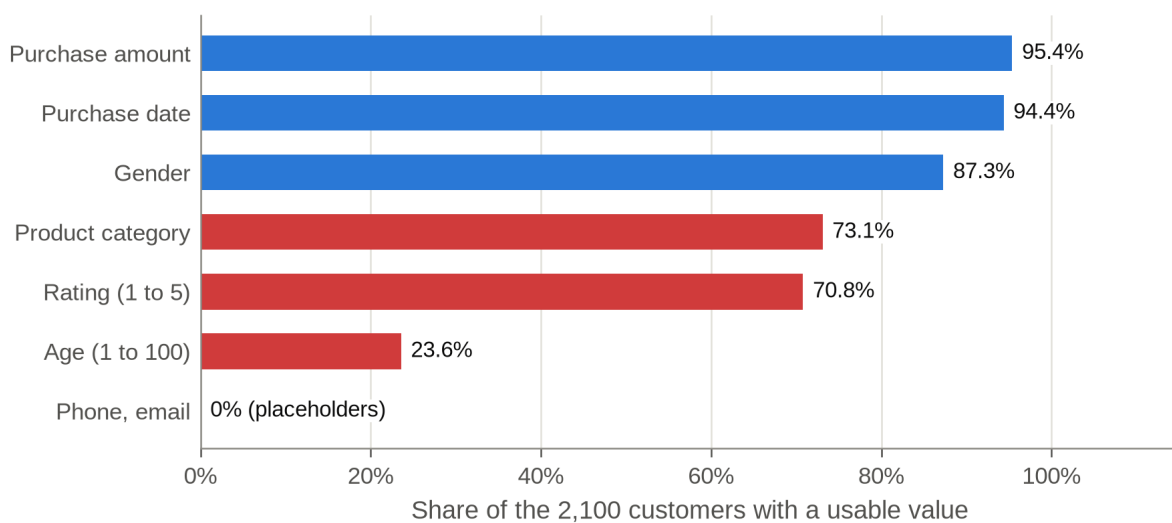


Figure 1: Share of the 2,100 customers with a usable value in each analytical column after cleaning. Columns below 80% are shown in red. Amount, date and gender are usable; category, rating and above all age are not complete enough for segment analysis, and the contact fields are placeholders.

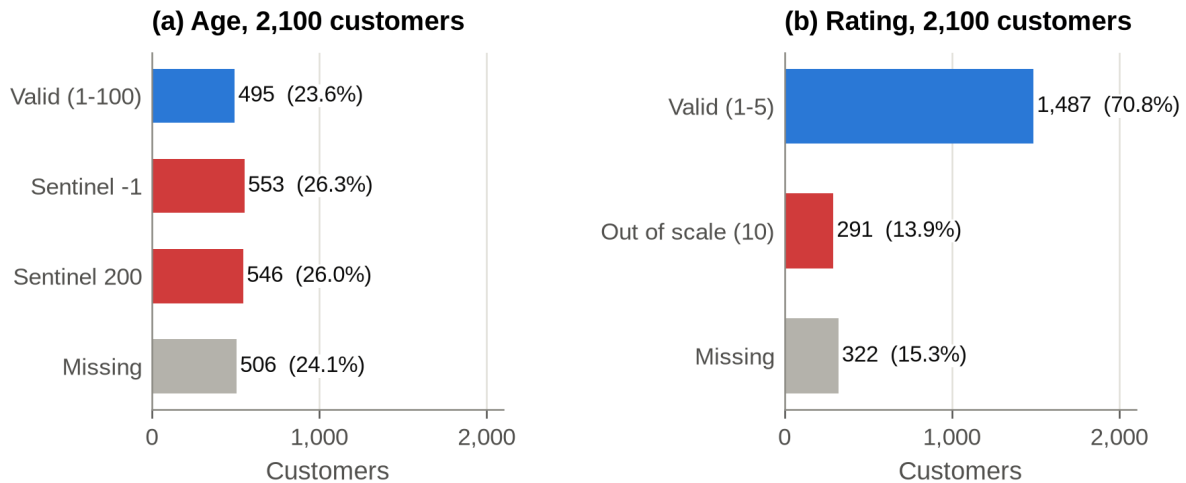


Figure 2: What the raw age (a) and rating (b) columns contain. The two age sentinels weigh as much as the valid ages and the missing values together; a mean on the raw column gives 85 years instead of 54.4. The 10s raise the raw mean rating from 3.03 to 4.17.

4 Cleaning

Table 2: One decision per problem, with the reason for it

Problem		Count	Decision and reason
Duplicate	Cus-tomerID	50	First row kept. Each duplicate is an exact copy of its first row, field by field, so no information is lost.
Export artefacts	2 columns		Unnamed (empty) and the copy of Gender dropped.
Gender spellings		1,833	Six spellings (M, male, Male, F, female, Female) mapped to Male and Female; 267 left missing.
Sentinel ages		1,099	Set to missing, not to a mean. An imputed age would invent a demographic profile for three customers in four.
Ratings of 10		291	Set to missing. They could be 1 or 5 on a reversed scale; neither can be proved.
Sentinel date		118	Flag <code>DateInvalid</code> , date left empty. The purchase, amount and rating remain usable.
Missing category		565	Labelled <code>Unknown</code> and kept, so that revenue totals remain complete.
Missing amount		97	Flag <code>AmountMissing</code> , kept missing.

Every flagged row is kept, so a reader can include or exclude it and see the difference. The clean file `data/customers_clean.csv` has 2,100 rows. The notebook's cleaning and the production pipeline `etl/run_etl.py` produce the same table, column by column.

5 Indicators

Table 3: Each indicator with the population it is computed on

Indicator	Value	Computed on
Customers	2,100	distinct CustomerID
Revenue	\$1,021,324	2,003 recorded amounts
Average ticket	\$509.90	2,003 recorded amounts
Median ticket	\$519.42	2,003 recorded amounts
Mean rating	3.03 / 5	1,487 valid ratings
Ratings of 1 or 2	38.7%	1,487 valid ratings
Male / female	51.4% / 48.6%	1,833 known genders
Mean age	54.4 years	495 valid ages
Unknown category	565 (26.9%)	all customers
Invalid dates	118 (5.6%)	all customers
Missing amounts	97 (4.6%)	all customers

Most indicators name their own base. Revenue rests on 2,003 amounts, the rating on 1,487 and the mean age on 495. Writing the base beside the number is better than a footnote, because a footnote does not travel when a figure is copied into a slide.

6 Results

6.1 What is bought

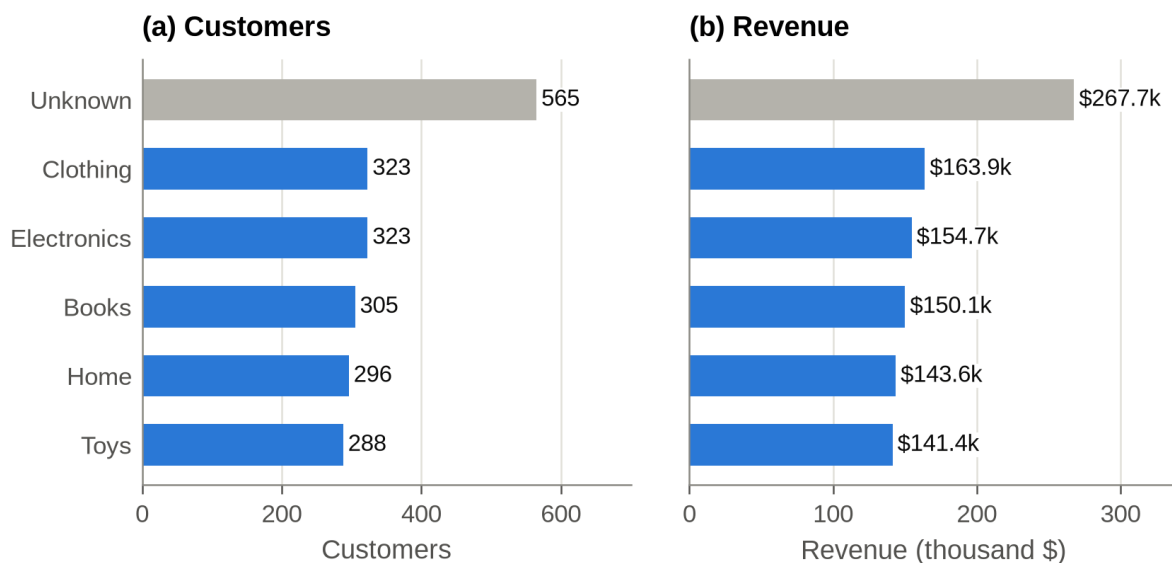


Figure 3: Customers (a) and revenue (b) by product category, in the same order. *Unknown* (grey) is a capture gap, larger than any real category. Among the five categories, Clothing leads revenue (\$163,865) and Toys trails (\$141,384), a gap of 14%.

The five real categories are close in size, from 288 (Toys) to 323 customers (Clothing and Electronics, tied). Because the *Unknown* purchases do not differ from the others in amount ($p = 0.21$), the shares of the known categories are a fair guide to the mix, but every category total is understated by about a quarter.

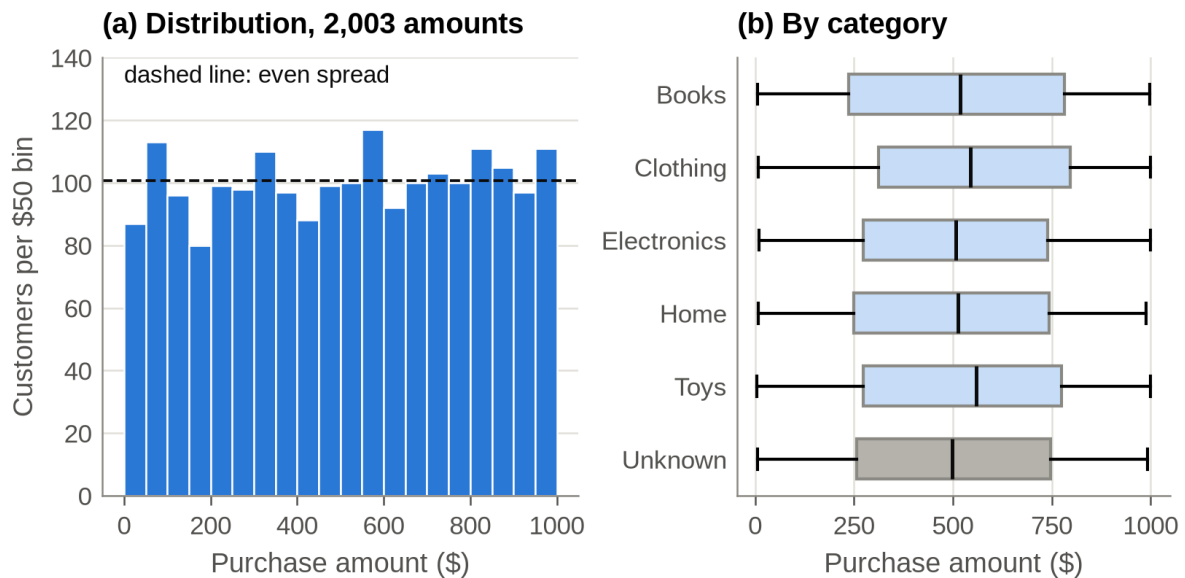


Figure 4: Distribution of the 2,003 recorded amounts in bins of \$50, against the count expected if amounts were spread evenly (dashed line) (a), and amounts by category (b). Boxes show the interquartile range and black lines the median. The distribution is flat from \$5 to \$1,000 (Kolmogorov–Smirnov $p = 0.49$), and the categories do not differ (Kruskal–Wallis $p = 0.45$).

6.2 Purchases through time

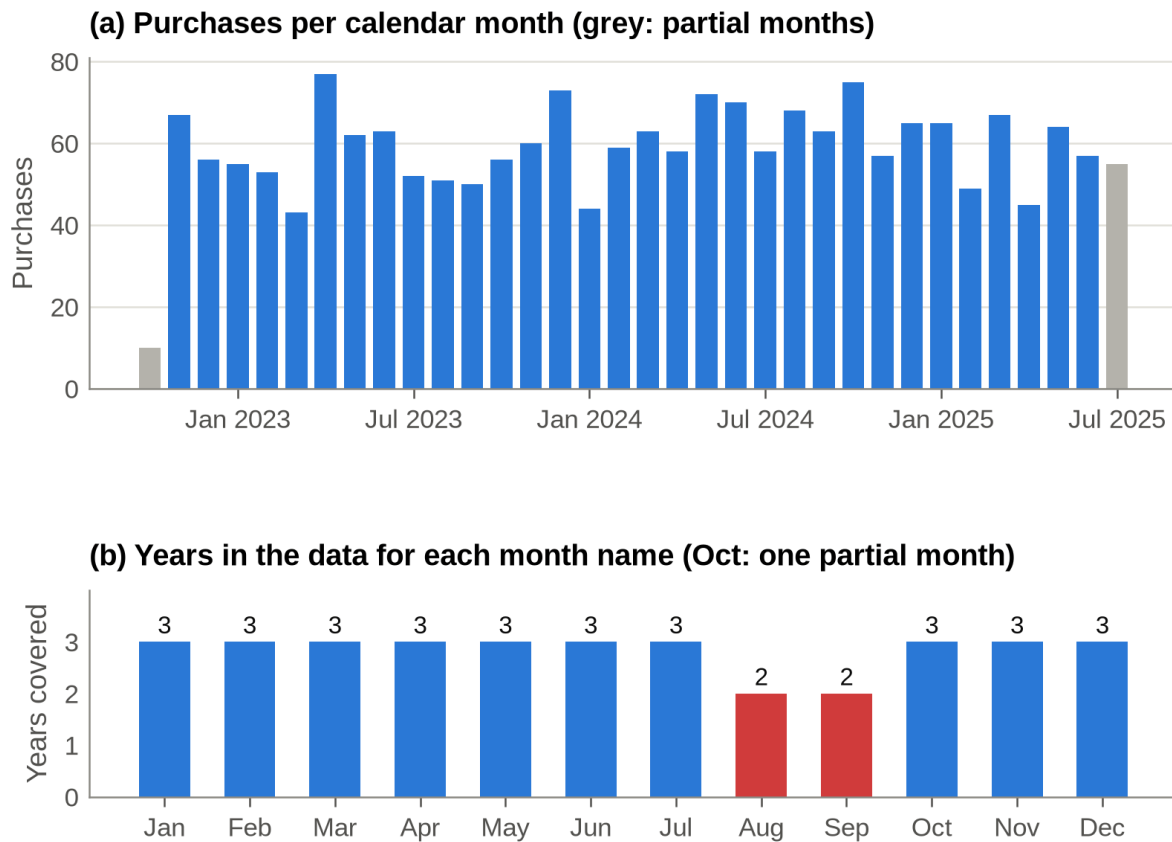


Figure 5: Purchases per calendar month from October 2022 to July 2025 (a); the first and last months are partial (grey). Number of years present in the data for each month name (b). August and September appear in two years, every other month in three. On the 32 complete months, purchases are compatible with an even spread (chi-square $p = 0.11$).

Table 4: The pooled monthly count divided by the number of years covered

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Pooled count	164	161	173	180	198	190	165	119	113	141	184	194
Years covered	3	3	3	3	3	3	3	2	2	3	3	3
Per year	54.7	53.7	57.7	60.0	66.0	63.3	55.0	59.5	56.5	47.0	61.3	64.7

Once divided by the years covered, August and September sit in the middle of the range. October is the lowest only because October 2022 holds the first five days of the period (10 purchases).

6.3 Who the customers are

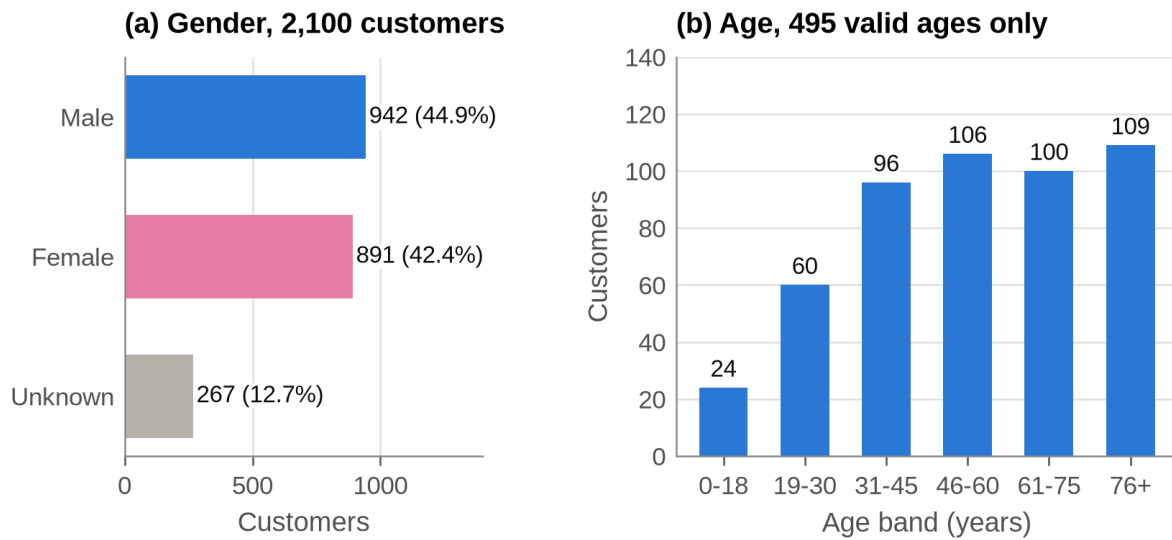


Figure 6: Gender of the 2,100 customers (a) and age bands of the 495 customers with a valid age (b). Where gender is known, the split is 51.4% male and 48.6% female. The age bands describe 23.6% of customers only and cannot be generalised.

Men and women do not differ in spending (Mann–Whitney $p = 0.10$): average ticket \$499 for men, \$522 for women.

6.4 Satisfaction

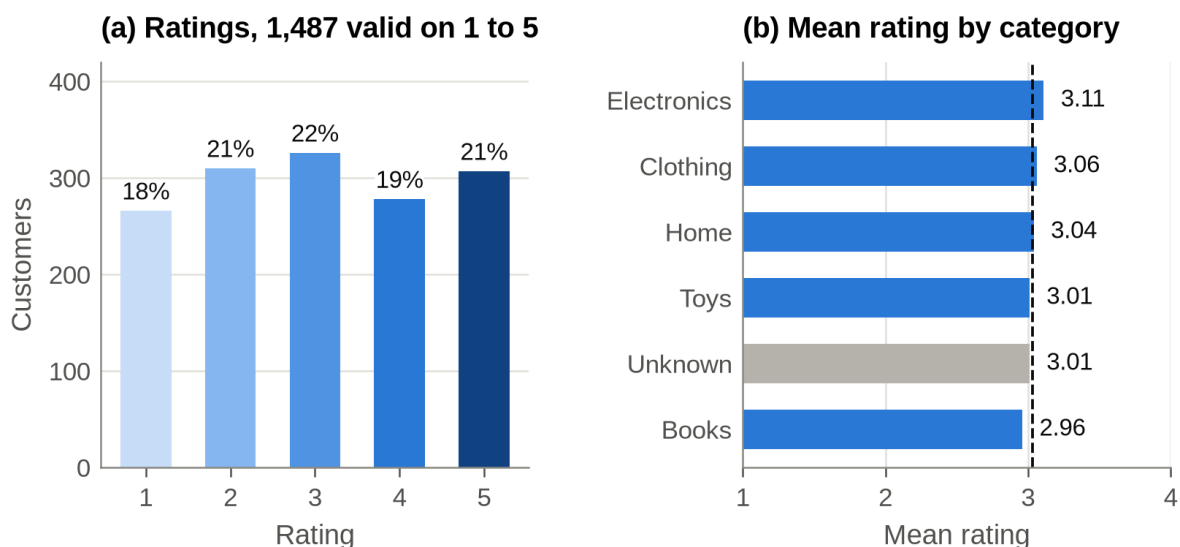


Figure 7: Distribution of the 1,487 valid ratings (a) and mean rating by category against the overall mean of 3.03 (dashed line) (b). The five scores are almost equally frequent (18% to 22%), and the rating mix does not depend on the category (chi-square $p = 0.94$).

6.5 What drives the amount

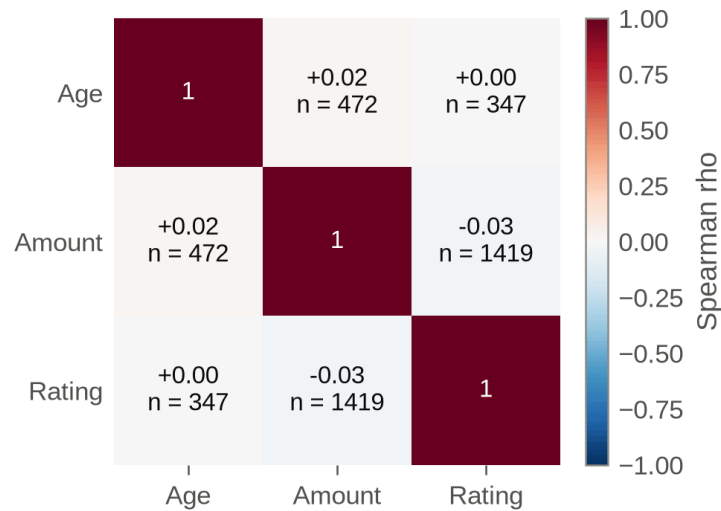


Figure 8: Spearman rank correlations between age, amount and rating, each on the customers where both values are valid (n in each cell). Every coefficient lies between -0.03 and $+0.02$: none of the recorded attributes is associated with spending.

7 Statistical tests

Table 5: Tests behind every segment statement of this report

Question	Test	Statistic	p	n
Amount differs by category	Kruskal–Wallis H	3.72	0.445	1,464
Amount differs, men against women	Mann–Whitney U	361,889	0.095	1,743
Amount differs, Unknown against others	Mann–Whitney U	380,118	0.209	2,003
Rating mix differs by category	χ^2 (16 df)	8.38	0.937	1,072
Purchases differ across 32 months	χ^2 (31 df)	40.98	0.108	1,917
Amount associated with rating	Spearman ρ	-0.03	0.301	1,419
Amount associated with age	Spearman ρ	0.02	0.702	472
Amounts spread evenly	Kolmogorov–Smirnov D	0.019	0.488	2,003

None of the eight tests is significant at the 5% level. This is itself the result: the register does not separate customers who spend more from those who spend less, and the monthly profile does not show a season. Uniform amounts, uniform ratings and 48 distinct names point to a synthetic or anonymised extract. The methods and the pipeline transfer unchanged to real data; the business conclusions must be re-tested on it.

8 Insights and recommendations

8.1 Data capture is the first problem to solve

Four of the six quality policies are breached: invalid dates (5.6% against a 3% threshold), unknown category (26.9% against 20%), missing age (76.4% against 30%) and missing rating (29.2% against 25%). The contact fields cannot reach a single customer.

Recommendation. Enforce validation at entry: age between 1 and 100, a rating widget limited to 1 to 5, a mandatory category, a calendar date picker, and unique, verified contact details. Keep the automated quality gate in the publishing path until the four policies pass. *Owner: point-of-sale and CRM IT.*

8.2 Revenue is \$1,021,324 on 2,003 recorded amounts

The average ticket is \$509.90, the median \$519.42. 97 amounts are missing, so revenue is a recorded total, not a complete one.

Recommendation. Publish revenue with its base and make the amount mandatory for closed transactions. *Owner: finance.*

8.3 The Unknown category is the largest “category”

565 purchases have no category, more than Clothing or Electronics (323 each). Their amounts do not differ from the others, so the known mix is a fair guide, but each category total is understated by about a quarter.

Recommendation. Make the category mandatory at the point of sale before any assortment decision. The 14% revenue gap between Clothing and Toys comes from volume, not from the ticket ($p = 0.45$). *Owner: merchandising.*

8.4 No customer attribute predicts spending

Amounts do not differ by gender, category, age or rating, and are spread evenly between \$5 and \$1,000.

Recommendation. Do not target campaigns on age, gender or category with this data: none separates high from low spenders. Collect purchase history, channel and basket content, the usual drivers of spend. *Owner: marketing.*

8.5 Satisfaction is mediocre and polarised

The mean rating is 3.03 out of 5. 38.7% of valid ratings are 1 or 2, in every category alike.

Recommendation. Contact customers who rate 1 or 2 within a week, and add a reason code to the rating to find the cause. *Owner: customer service.*

8.6 The monthly profile shows no seasonality

The dip of August and September in the pooled chart is a coverage effect. On the true monthly series, purchases are even ($p = 0.11$).

Recommendation. Do not plan stock or staff on the pooled chart. Report purchases per calendar month, and re-test seasonality once three full years are available. *Owner: operations.*

9 The interactive dashboard

`dashboard.html` is a single self-contained file built with Plotly: open it in a modern browser (Chrome or Edge, then Ctrl+F5), with no server and no installation. The left navigation bar opens nine views: Dashboard, Data Explorer, Live Data, Customers, Products, Trends, Data Quality, Insights and KPI Code. Filters on category, gender, age band and rating update every key indicator, chart and table at once. The header carries a live clock, an auto-refresh selector and the data-quality status; the side panel offers dark and light themes, six accent colours and four fonts. The Data Explorer exports the filtered table to CSV or Excel.

The dashboard is reproduced here without any modification. The captures were taken with every filter cleared, at 1,920 by 1,057 pixels.

9.1 Reading the dashboard against this report

Every headline tile of the dashboard equals the report value: 2,100 customers, \$1,021,324 revenue, \$510 average ticket, 3.03 mean rating, 54.4 mean age, 50 duplicates removed and 118 invalid dates. Four elements use a different base or rule; they are listed so that no reader meets an unexplained difference.

Table 6: Reconciliation between dashboard elements and report indicators

Dashboard element	Base used and report equivalent
<i>Gender mix donut</i>	All 2,100 customers: Male 44.9%, Female 42.4%, Unknown 12.7%. Report equivalent: 51.4% / 48.6% of the 1,833 known genders.
<i>Purchases over time, Monthly purchase trend, Dynamic analytics</i>	Three years pooled by month name. August and September count two years, the other months three. Report equivalent: Figure 5 and Table 4.
<i>Avg purchase by category (Products)</i>	Revenue divided by all customers of the category, including the 97 without an amount (Clothing \$507). Report equivalent: mean of recorded amounts (Clothing \$537).
<i>Data quality control alerts</i>	One high, two medium and one low alert: Unknown category graded medium, rating sparsity low. Pipeline policy (<code>results/quality_alerts.json</code>): two high, two medium. Both give the status ALERT.

The Data Explorer and Customers tables show the purchase date as captured, so the 118 flagged rows display 32/13/2020; they are excluded from every time-based chart.

9.2 Dashboard

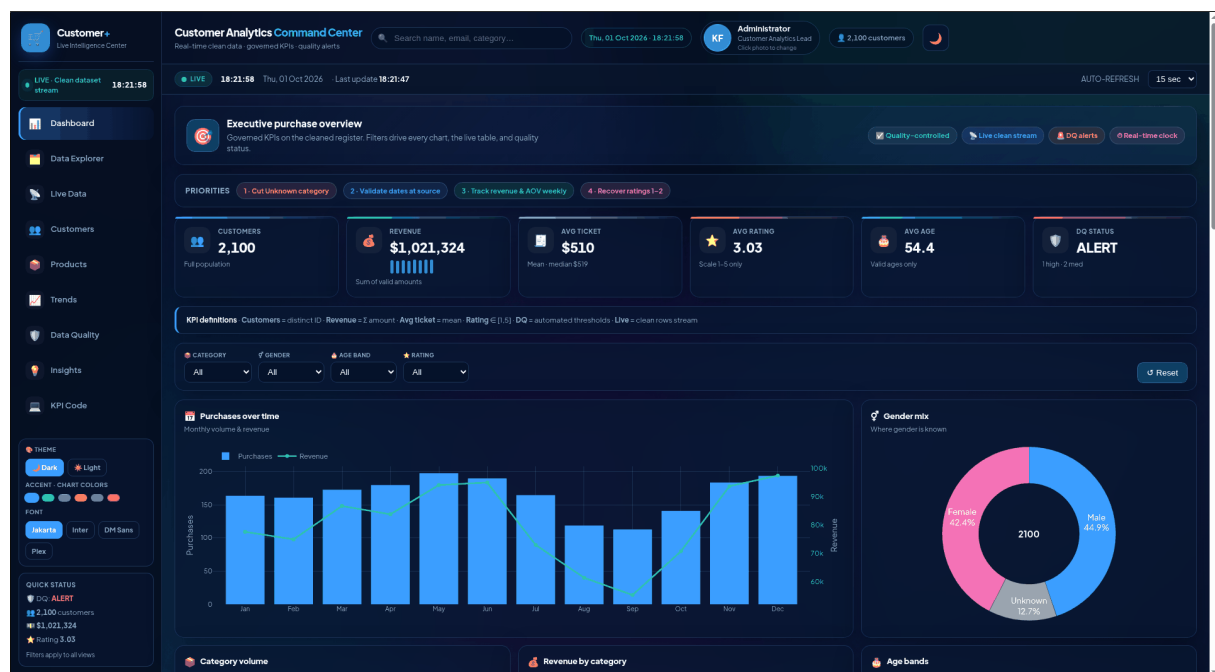


Figure 9: Dashboard, header and executive overview: live clock, priorities, the six key indicators (customers, revenue, average ticket, rating, age, data-quality status), KPI definitions, filters, purchases over time and gender mix.



Figure 10: Dashboard, category volume, revenue by category, age bands (the 1,605 customers without a valid age shown as *Unknown*), rating distribution and key insights.



Figure 11: Dashboard, dynamic analytics (bars, lines or area; revenue, volume or average ticket by month), live category pulse and the rolling window of the clean-data stream.



Figure 12: Dashboard, interactive category treemap (a click filters every view), revenue against volume by category with the mean rating as colour, and the live KPI pulse.

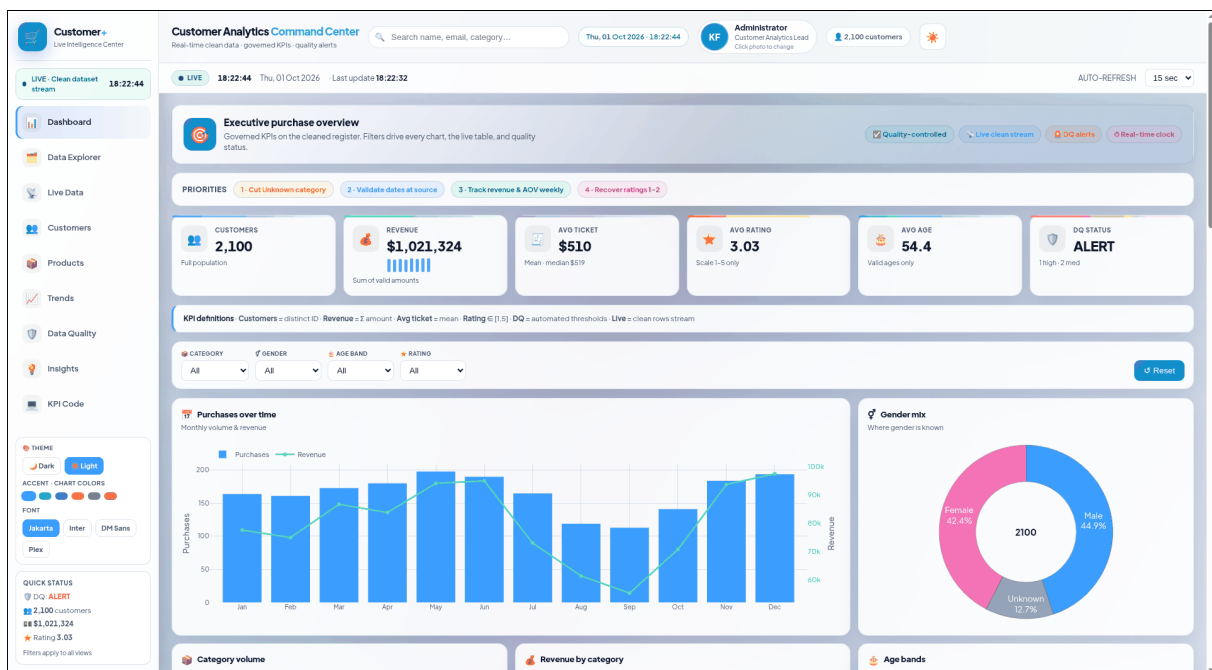


Figure 13: Dashboard in the light theme.

9.3 Data Explorer and Live Data

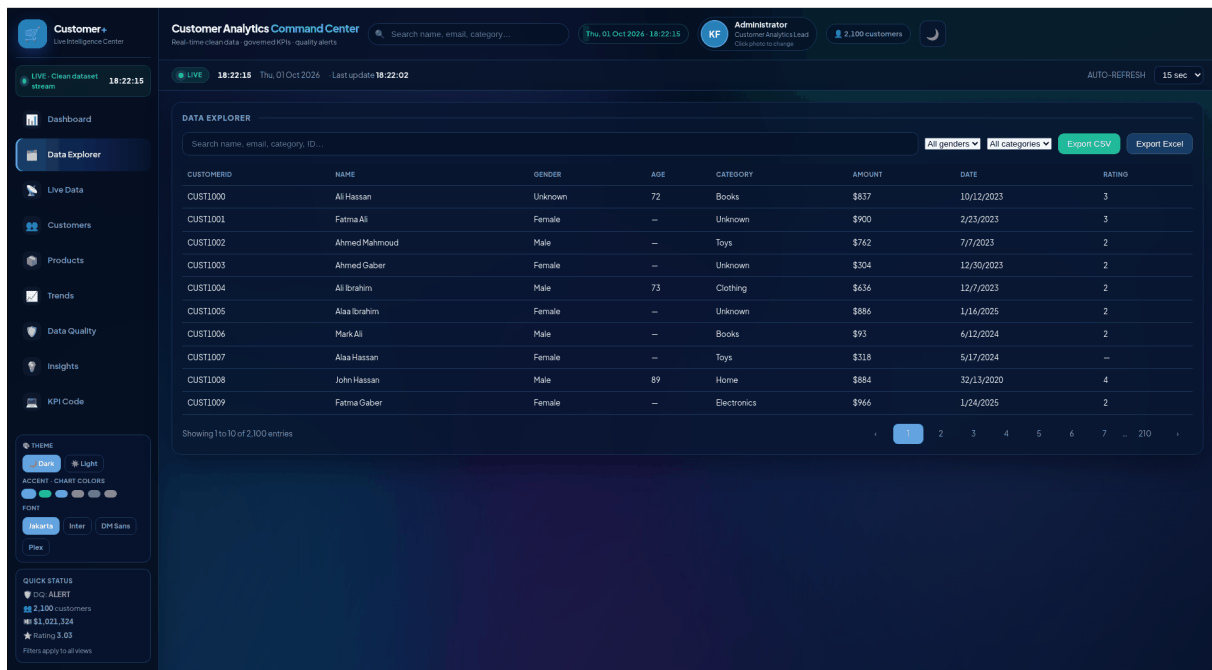


Figure 14: Data Explorer: search, gender and category filters, CSV and Excel export, paginated over the 2,100 customers.

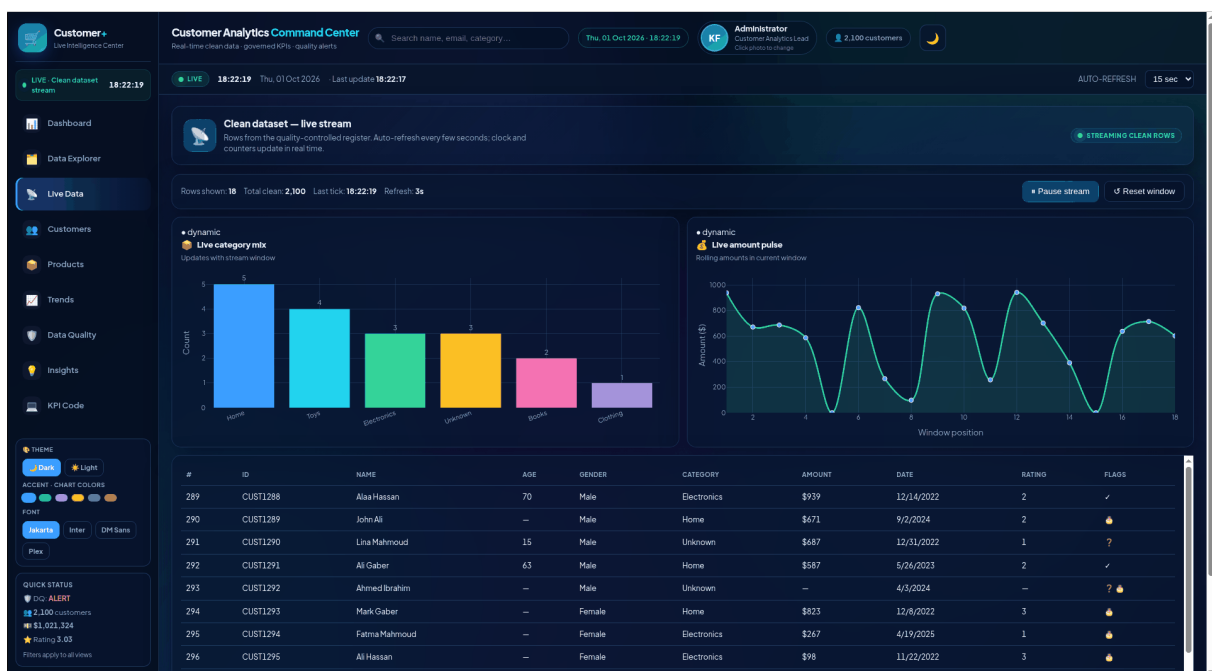


Figure 15: Live Data: the clean register streamed row by row, with the live category mix, the rolling purchase amount and the quality flags of each row.

9.4 Customers and Products

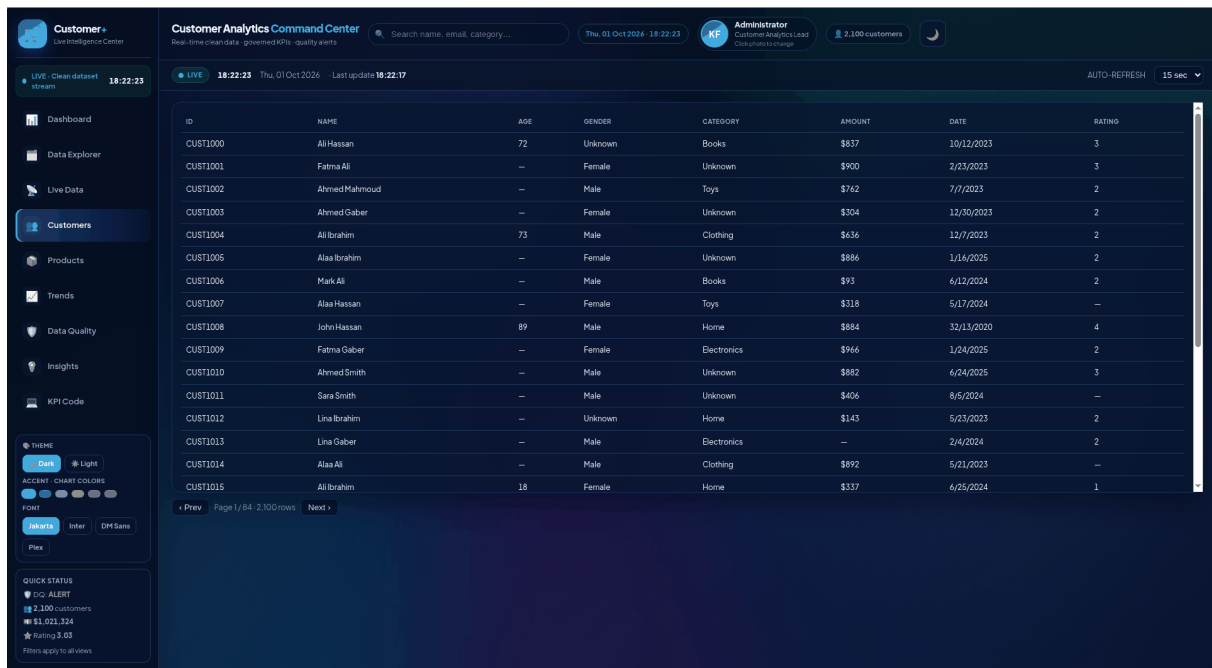


Figure 16: Customers: one row per customer with age, gender, category, amount, date and rating.

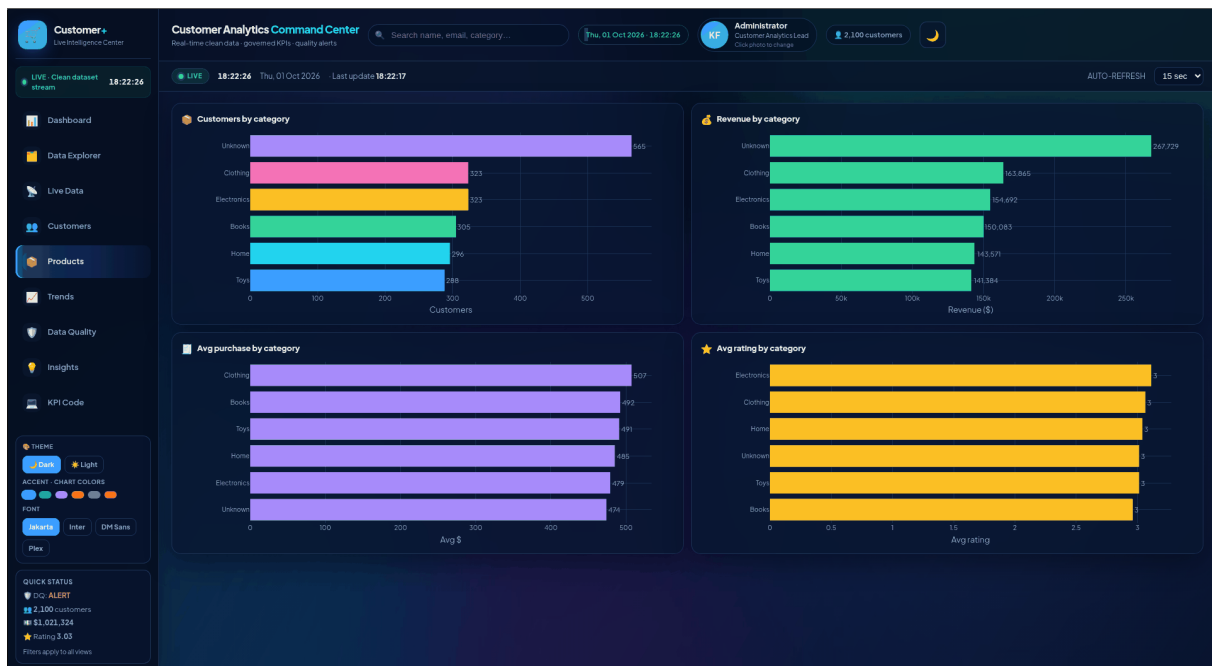


Figure 17: Products: customers, revenue, average purchase and mean rating by category.

9.5 Trends



Figure 18: Trends: monthly purchase trend (pooled by month name), purchase amount distribution and rating mix.

9.6 Data Quality

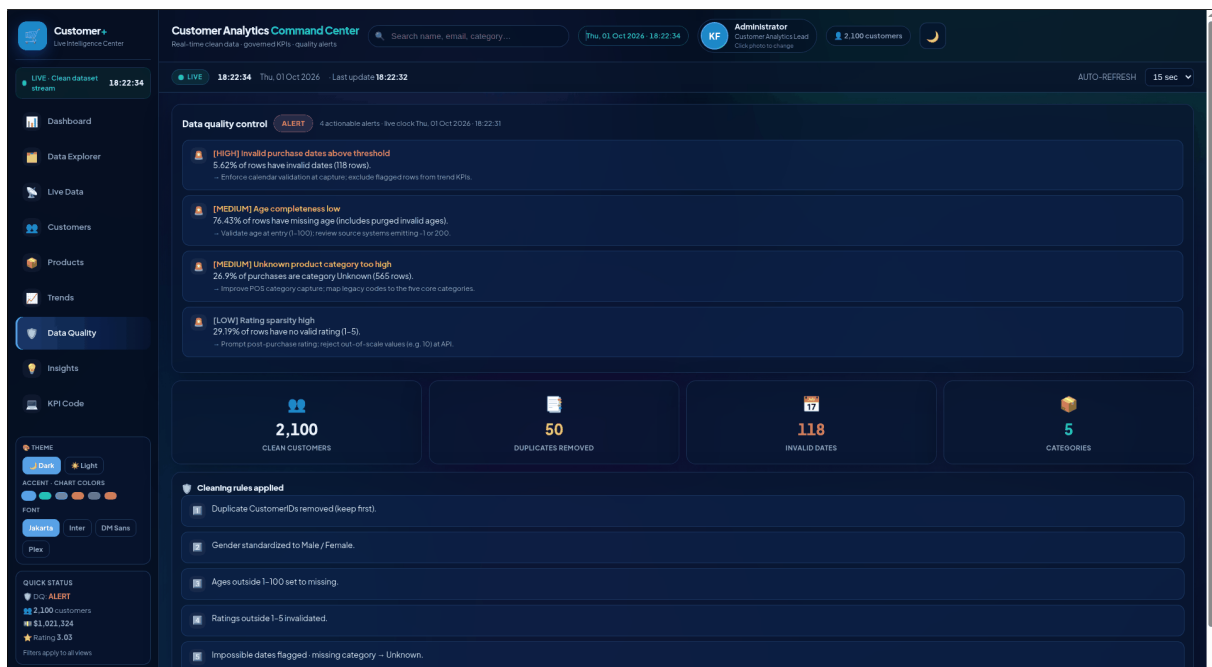


Figure 19: Data Quality: the four actionable alerts with their recommendations, the defect counts (2,100 clean customers, 50 duplicates removed, 118 invalid dates, 5 categories) and the cleaning rules applied.

9.7 Insights and KPI Code

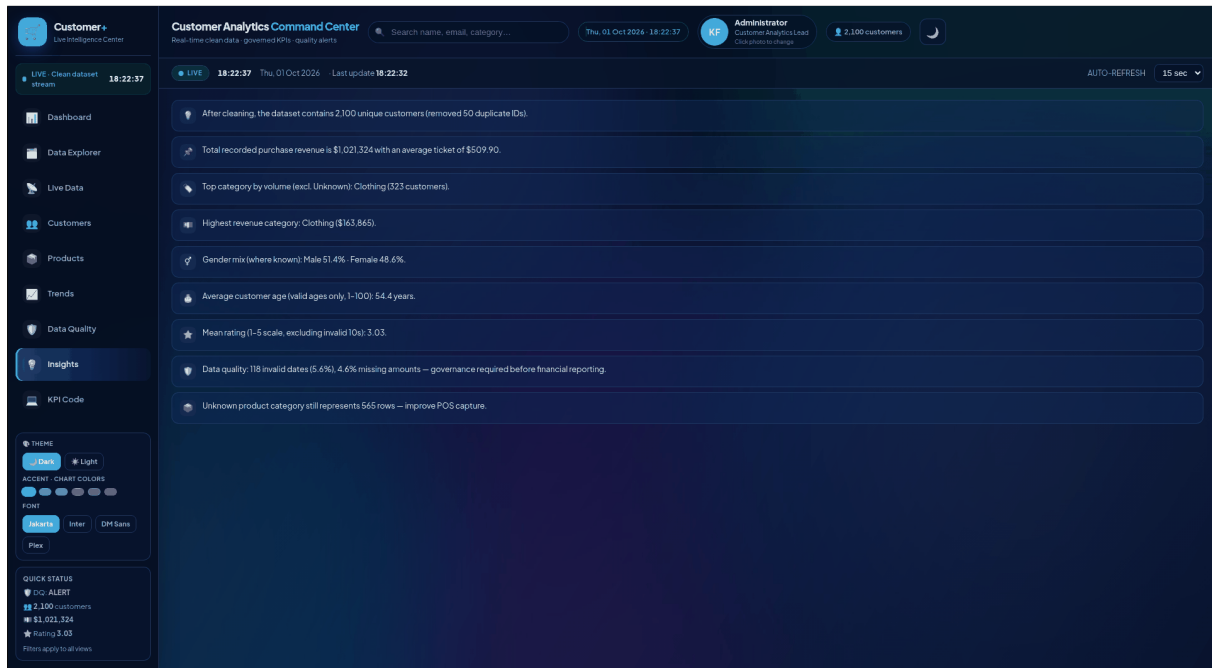


Figure 20: Insights: the nine headline findings computed from the clean register.

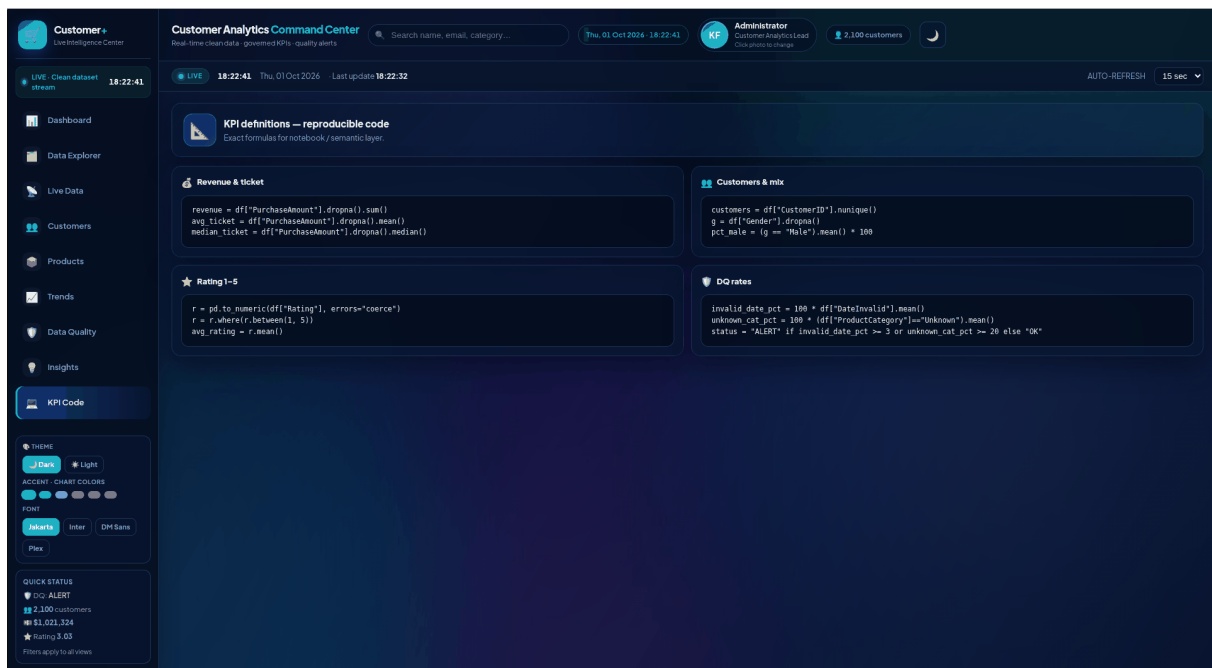


Figure 21: KPI Code: the exact pandas formula behind each indicator (revenue and ticket, customers and mix, rating, data-quality rates), so that every tile can be reproduced.

10 Pipeline and automated quality alerts

The dashboard is only as reliable as the data it reads. The project therefore ships the pipeline that produces and guards the clean register.

Table 7: Components of the pipeline

Component	Role
<code>etl/run_etl.py</code>	Extract the raw file, apply the cleaning rules, load <code>data/customers_clean.csv</code> , write an audit (<code>results/etl_audit.json</code>).
<code>api/run_quality_alerts.py</code>	Compare the clean table with the policy thresholds; exit code 0 (OK), 1 (WARN) or 2 (ALERT).
<code>api/notify_alerts.py</code>	Send the alerts to the console, a file, a webhook or email (<code>api/notify_config.json</code>).
<code>api/metrics_server.py</code>	Expose the indicators and alerts to Prometheus.
<code>airflow/customer_analytics_etl_dag.py</code>	Run daily: ETL, then the quality gate, then publication; an ALERT stops publication.
<code>grafana/, docker-compose.yml</code>	Prometheus and Grafana monitoring of the indicators and alerts.

Table 8: Quality policies applied by the pipeline to the clean register

Policy	Value	Threshold	Severity	Breached
Invalid purchase dates	5.62%	3%	high	yes
Unknown product category	26.90%	20%	high	yes
Missing purchase amount	4.62%	5%	high	no
Missing age after purge	76.43%	30%	medium	yes
Missing or invalid rating	29.19%	25%	medium	yes
Duplicate CustomerID	0	1	medium	no

The status is **ALERT**: two high-severity policies are breached. The duplicate policy reads 0 because it is evaluated on the clean table, after the 50 duplicates were removed; the ETL audit records the 50 removals. Missing amounts sit just below their threshold and should be watched weekly.

11 Limitations

One purchase per customer. No repeat-purchase, retention or lifetime-value indicator can be computed.

Age rests on 495 customers, 23.6% of the file, and nothing shows that they represent the others. Age bands are descriptive only.

Ratings rest on 1,487 customers, 70.8% of the file. The 291 values of 10 may hide real scores typed on another scale.

The extract looks synthetic. 48 distinct names, two phone numbers, 144 emails, and amounts and ratings spread evenly. Methods and pipeline transfer to real data; the business conclusions must be re-tested on it.

Association, not causation. The absence of a relationship between spend and the attributes recorded does not exclude drivers that are not recorded.

12 Reproducibility

Every number, table and figure of this report is produced by the notebook `Customer-Analytics.ipynb` from the raw file `data/customers_raw.csv`. The notebook writes the clean register (`data/customers_clean.csv`), the result tables (`results/`) and the figures (`figures/`, PNG at 300 dpi), and checks that its cleaning matches the production ETL. Once the capture is repaired at source, re-running the notebook and the pipeline unchanged recomputes every flag, indicator, test, figure and alert.

KOUAME Koffi Fidèle
Data Analysis Internship
koffifidelek59@gmail.com